

Installation Guide

Project: Smart Course Companion

Course: SOEN 287 – Web Programming

Team: TeamA

1. Overview

This guide explains how to install and run the **Smart Course Companion** web application locally.

The project consists of:

- Frontend (HTML, CSS, JavaScript)
- Backend (Node.js with Express)
- Database (MongoDB)

2. Prerequisites

Ensure the following are installed on your system:

- **Node.js** (v16 or higher recommended)
- **npm** (comes with Node.js)
- **MongoDB** (local installation or MongoDB Atlas)
- A modern web browser (Google Chrome recommended)

3. Download and Extract the Project

1. Download the project .zip file from Moodle
2. Extract the files:
 - Right-click → **Extract All** (Windows)
 - Or unzip (Mac)

4. Project Structure Overview

After extraction, the project contains:

SmartCourseCompanion/

```
|
|
|— HTML files (e.g., home-page.html, dashboard.html)
|
|— css/          # Stylesheets
|
|— js/           # Frontend scripts (e.g., API.js)
|
|— images/       # Images used in the UI
|
|
|— backend/
|   |— server.js
|   |— package.json
|   |— package-lock.json
|   |— .env.example
|
|
|   |— config/
|   |   |— db.js
|
|
|   |— middleware/
|   |   |— auth.js
|   |   |— authMiddleware.js
|   |   |— upload.js
|
```

```
| |— models/
| | |— Assessment.js
| | |— Course.js
| | |— CourseTemplate.js
| | |— Submission.js
| | |— User.js
|
| |— routes/
| | |— admin.js
| | |— assessments.js
| | |— auth.js
| | |— courses.js
| | |— submissions.js
```

5. Navigate to Backend Folder

Open a terminal and move to the backend directory:

```
cd SmartCourseCompanion/backend
```

6. Install Dependencies

Run the following command:

```
npm install
```

This installs all required Node.js packages listed in *package.json*.

7. Environment Configuration

The project includes:

```
.env.example
```

Steps:

1. Create a copy of the file
2. Rename it to:


```
.env
```
3. Open `.env` and configure:

Example:

```
MONGO_URI=mongodb://localhost:27017/smartcourse
```

```
PORT=3000
```

If using MongoDB Atlas, replace `MONGO_URI` with your cluster connection string.

8. Start MongoDB

Option 1: Local MongoDB

Make sure MongoDB is running on your system.

Option 2: MongoDB Atlas

No local setup required — ensure the connection string is correct.

9. Run the Backend Server

In the terminal (inside backend folder), run:

```
node server.js
```

Alternatively:

```
npm start
```

10. Access the Application

1. Open a web browser

2. Go to:

```
http://localhost:3000
```

3. Use the interface to:

- Sign up / log in
- Access dashboards
- Manage courses and assessments

11. Notes

- Backend logic is organized as follows:
 - routes/ → handles API endpoints
 - models/ → defines database structure
 - middleware/ → handles authentication and file uploads
 - config/db.js → manages database connection
- Frontend communicates with backend using JavaScript (e.g., js/API.js)
- All data is stored in MongoDB

12. Troubleshooting

Server does not start

- Ensure you are inside the backend folder:

cd backend

- Run:

node server.js

Database connection error

- Check .env file values
- Ensure MongoDB is running
- Verify connection string

npm install fails

npm cache clean --force

npm install